

# Companion LXXX: Neural Schrödinger Flows and Score-Transport Geometry of Persistence Diagrams in the Relaxation-Driven Cyclic Cosmology

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## Abstract

Neural Schrödinger flows unify score-based diffusion models and Schrödinger bridge dynamics into a single framework for learning stochastic transport processes. Applied to persistence diagrams, these flows learn the time-dependent drift and diffusion fields that interpolate between topological ensembles in a probabilistic and geometrically meaningful way. Building on the entropic optimal transport and score-based models developed in previous Companions, this work introduces neural Schrödinger flows for persistence diagrams in the Relaxation-Driven Cyclic Cosmology (RDCC). The relaxation mechanism modifies the scale-dependent evolution of the gravitational potential, inducing characteristic changes in score fields, transport geometry, and stochastic flow trajectories. The resulting framework provides a powerful and flexible tool for modeling RDCC signatures in weak-lensing topology.

## 1 Introduction

Persistence diagrams encode the birth and death of topological features in weak-lensing convergence fields. Previous Companions introduced Wasserstein geometry, entropic optimal transport, Schrödinger bridges, and score-based diffusion models as tools for analyzing the Relaxation-Driven Cyclic Cosmology (RDCC).

Neural Schrödinger flows unify these approaches by learning the drift and diffusion fields of a stochastic differential equation (SDE) that transports one distribution of persistence diagrams into another. This combines the dynamic interpretation of Schrödinger bridges with the generative power of score-based models, yielding a flexible and expressive framework for modeling topological distributions.

In RDCC, the relaxation mechanism modifies the underlying gravitational potential in a scale-dependent manner, producing characteristic signatures in the learned transport geometry.

## 2 Neural Schrödinger Flows

A neural Schrödinger flow is defined by the SDE

$$dD_t = v_\theta(D_t, t) dt + \sqrt{2\varepsilon(t)} dB_t, \tag{1}$$

where

- $v_\theta(D_t, t)$  is a neural network approximating the drift,
- $\varepsilon(t)$  is a time-dependent diffusion schedule,

- $B_t$  is Brownian motion on diagram space.

The flow is trained to match boundary distributions  $\rho_0$  and  $\rho_1$  by minimizing a Schrödinger-bridge-inspired objective.

### 3 Score-Transport Geometry

The drift is decomposed as

$$v_\theta(D, t) = -\varepsilon(t) s_\theta(D, t) + u_\theta(D, t), \quad (2)$$

where

- $s_\theta(D, t)$  is the learned score field,
- $u_\theta(D, t)$  is a learned residual transport term.

This decomposition yields a geometric interpretation:

- the score term drives the flow toward high-density regions,
- the residual term encodes transport geometry beyond pure diffusion.

### 4 RDCC Modification of Neural Schrödinger Dynamics

In RDCC, the convergence evolves as

$$\kappa_{\text{RDCC}}(\ell) = \kappa(\ell) \left[ 1 - \chi_\kappa \left( \frac{\ell}{\ell_{\text{rel}}} \right)^\alpha \right]. \quad (3)$$

This modifies the log-density of persistence diagrams:

$$\log p_{\text{RDCC}}(D) = \log p(D) - \chi_p \left( \frac{\ell}{\ell_{\text{rel}}} \right)^\alpha. \quad (4)$$

The RDCC score becomes

$$s_{\text{RDCC}}(D, t) = s(D, t) \left[ 1 - \chi_s \left( \frac{\ell}{\ell_{\text{rel}}} \right)^\alpha \right]. \quad (5)$$

The RDCC drift is

$$v_\theta^{\text{RDCC}}(D, t) = v_\theta(D, t) \left[ 1 - \chi_v \left( \frac{\ell}{\ell_{\text{rel}}} \right)^\alpha \right]. \quad (6)$$

### 5 RDCC Signatures in Neural Transport Geometry

RDCC induces:

- reduced score magnitude at small scales,
- enhanced drift at large scales,
- modified stochastic flow trajectories,
- altered dynamic barycenters,
- scale-dependent deformation of transport geometry,
- characteristic curvature in learned Schrödinger flows.

These signatures provide a generative and dynamic probe of the RDCC gravitational field.

## 6 Conclusion

Neural Schrödinger flows unify score-based diffusion models and Schrödinger bridge dynamics into a single framework for learning stochastic transport processes. In the Relaxation-Driven Cyclic Cosmology, the relaxation mechanism modifies the score field, drift dynamics, and transport geometry in a scale-dependent manner, producing distinctive signatures in weak-lensing topology. The present Companion establishes the foundation for future studies of neural SDEs, generative modeling, and dynamic geometric analysis in cosmological topology.

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